



Noroff

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and digital media

Technical Report

Semester Project 1

Ada Linn Taule

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1. Summary

This report presents the development of a responsive and interactive website for the new opening museum, *the Community Science Museum*. This is created as part of the Frontend Development course. The main purpose of this project was to design and implement a visually appealing, accessible, and user-friendly website that communicates the museum's mission to inspire curiosity and learning in science. The website aims to attract children, families, teachers, and researchers by providing information about exhibitions, events, and educational programs in an engaging and organized way.

The development process was divided into several phases, including research, design, coding and testing. During planning phase, a Kanban board was used to structure and manage tasks throughout the project. This approach made it easier to visualize progress, prioritize tasks, and maintain an efficient workflow by dividing activities into columns, such as *Backlog*, *To Do*, *In Progress*, *In Review*, and *Done*. The design process began with wireframing and low-fidelity prototypes to establish layout and content hierarchy. A color palette and typography were selected to reflect the museum's purpose and identity and maintain consistency across pages.

The website was built using semantic HTML for structure, CSS (mostly Flexbox) for layout and styling, JavaScript for dynamic elements such as responsive menu. Meida Query was used to create a responsive design. Accessibility principles were followed to improve readability and usability for all users. Tools such as Visual Studio Code, GitHub, and Figma supported the development and design process.

Through this project, I gained a deeper understanding of frontend development principles, responsive design and project organization using Kanban. And the final project is a responsive and informative website that fulfils the intended goals.



2. Body

2.1. Introduction

This project was developed as part of the Frontend Development course and focused on designing and implementing a responsive and user-friendly website for the upcoming new opening museum, the Community Science Museum. The aim of the project was to combine creative design with technical execution to produce a fully functional website that communicated the museum's mission to inspire curiosity and learning science.

The website targets three main groups of users who want to learn science through interactive educational content: primary and middle school children aged 7-15, families with young children, and teachers for educational purposes. The website offers simple access to exhibition details and event information and learning activities while providing an accessible and visually appealing design. The project serves as an opportunity to develop practical skills in designing user interface and creating responsive layouts and accessible web pages.

The development process started with research and planning followed by implementation of Kanban board to handle tasks effectively. The Kanban system enabled users to monitor their work progress while managing tasks priorities between Backlog, To Do, In Progress, In Review and done stages.

The design phase involved creating wireframe and visual mockups, using Canva for low-fidelity and Figma for high-fidelity, focusing on using a consistent color palette, clear typography and flexible layout for different screen dimensions. HTML and CSS with Flexbox were used during the implementation phase, JavaScript was used for responsive menu, and Media Query was used to create to achieve website's responsiveness. The development process used VSC and GitHub and browser developer tools for coding and collaboration and testing process.

This report outlines the entire development process, including planning, design decision, coding works, testing process and evaluation. The report also reflects on the difficulties together with important knowledge gained during the development process.



2.2. Main section of report

2.2.1 Planning Process

The planning stage was a critical foundation for the entire project. It provided structure and clarified, ensuring that every subsequent phase was guided by week-defined objectives. As mentioned earlier, the main goal of the project was to develop a responsive and educational website for the Community Science Museum – a digital space where children, families and teachers could explore scientific topics, as well as information about the museum itself.

This phase began with identifying the target audience, purpose of the website, and outline the site's information architecture. A sitemap was created to organize the main sections: Home (logo), Exhibitions, For Kids, For Teachers, Visit Us, Get Involved, and Contact. This ensured logical navigation and a user-friendly experience. Design principles such as simplicity, consistency, and accessibility were established early to guide visual and functional decisions.

To manage tasks effectively, an Agile-inspired approach was adopted using Kanban board. This visual tool supported flexibility and continuous improvement while maintaining transparency. The Kanban board was divided into five columns:

1. Backlog: contained all ideas and planned features that were yet to be started
2. To Do: Listed tasks that were ready to be worked on
3. In Progress: Ongoing tasks
4. In Review: completed tasks awaiting to be testing, validation, or feedback
5. Done: fully finished and got approved

2.2.2 Research and Design

Before the design process began, research was essential to understand what content and features needed to be included on the website. Since the site was intended to be informative and primarily aimed at children, careful consideration was given to how information would be presented in an engaging and accessible way. Several existing science museum websites served as inspiration, including VilVite, Teknisk Museum, and the Museum of Science in Norway. This research provided valuable insights into best practices for layout, color usage, and content structure, which guided the design decisions throughout the project.



To establish the visual identity, a moodboard was developed, incorporating color inspirations, typography ideas, and imagery that reflected the museum's core value of curiosity, discovery, and learning. The color palette was carefully selected to reflect the museum's mission and create an engaging experience for visitors. *Coral Red* was chosen as the primary color because of its strong visual impact and association with energy, excitement and curiosity: qualities that align with scientific exploration.



Figure 1: Color palette

To ensure readability and a clean aesthetic, white was used as the background color, creating a sense of openness and simplicity. Black was selected for body text, to provide a maximum contrast and legibility against the white background.

Typography was selected to balance readability with a sense of personality and creativity. *Poppins* was chosen for headings because of its modern style, which adds visual impact and complements the playful tone of the design. For body text, *Inter* was selected due to its clean, highly legible structure, making it ideal for longer passages and ensuring accessibility across devices.

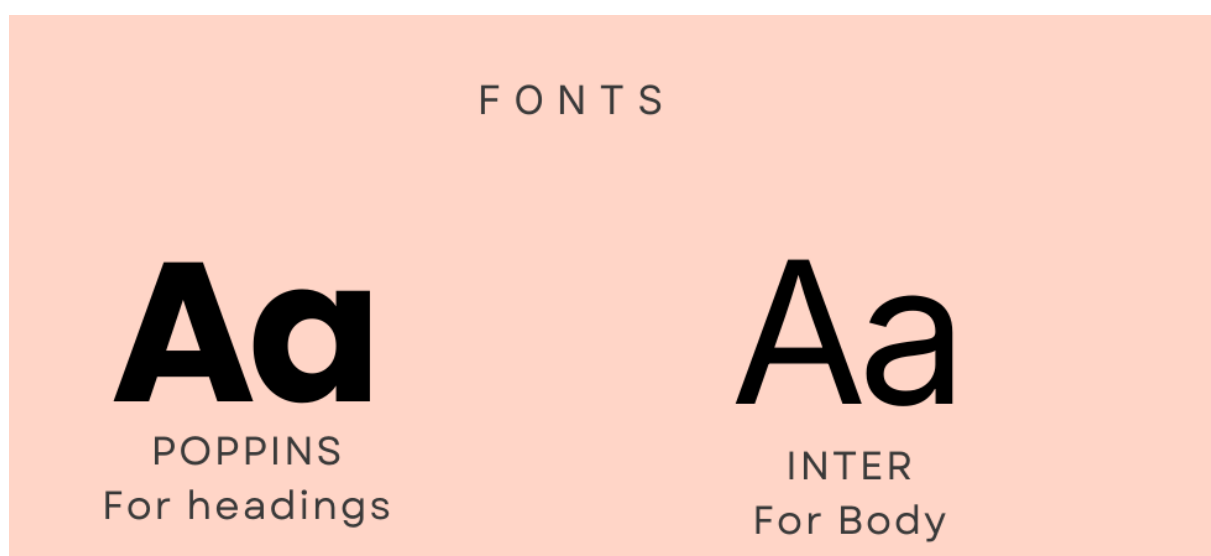


Figure 2: Typography example for website



The design process included two key prototyping stages to transition from concept to a realistic representation of the final website. A low-fidelity prototype was first created in Canva, focusing on layout and structure without color or typography. Using plain white and grey elements allowed to concentrate on content hierarchy and navigation flow without visual distractions.

Building on this foundation, a high-fidelity wireframe was developed in Figma. At this stage, the selected color palette, typography choices, and interactive elements were incorporated. The goal was to create a design that closely resembled the final website, as well as website's navigation was applied during this step, ensuring consistency and usability before moving into code development.

The homepage was designed as an introduction to the museum, featuring a hero image, a short welcome message, and buttons to separated pages. The Exhibitions page was structured to allow users to browse through topics, while Visit Us page provided practical information such as opening house, address, and ticket prices. For Kids and For Teachers pages were tailored to their specific audiences, featuring information content. Special Events page provides upcoming events that will take place in the museum, including information, date and time, for those who are interested.

2.2.3 Code Development

The development phase focuses on building a responsive, accessible, and interactive website using modern web technologies. This step is to bring High-fidelity wireframe, which was an early step, into real functional website. The foundation was established with a well-structure HTML layout, employing semantic elements such as `<header>`, `<main>`, `<section>`, and `<footer>`. The header contained the website logo and navigation menu, using a simple unordered list `<ul>` and `<li>`. The main section included content blocks that varied from page to page. The footer contained contact information, address, and social media.

Each page followed a consistent layout pattern to maintain familiarity across the site. Internal links were used for smooth navigation.

Flexbox was mainly used in this project, and media queries to adjust style for tablets, desktops and mobile screen. This ensure that the layout and visual hierarchy remained consistent across all devices. CSS variables (`--color-primary`, `--color-secondary`, `--h3-font-size`, and more) were used for easier theme management and to maintain consistency



throughout the website. The use of sufficient contrast also ensured compliance with accessibility guidelines (WCAG).

JavaScript was used to improve user interaction and engagement. One of the main scripts controlled the responsive navigation menu, allowing it to toggle visibility on smaller screens via a hamburger icon.

2.2.4 Testing and Evaluation

Testing Process started right away after code development finished. To verify the technical quality of the code, *W3C Markup Validation Service* was used to test the code, both HTML and CSS. The tool is used to detect syntax error, improper elements. Most of the pages passed validation without any major issues. However, the validation flagged a few minor errors related to images filenames, where spaces had been used between words (for example, `science museum.png`). The images were exported directly from Figma, which is the reason why there were spaces. But according to HTML standards, spaces in file names are not allowed, therefore I replaced them with **hyphens** (-). The CSS codes were also validated, and there were no errors to be found.

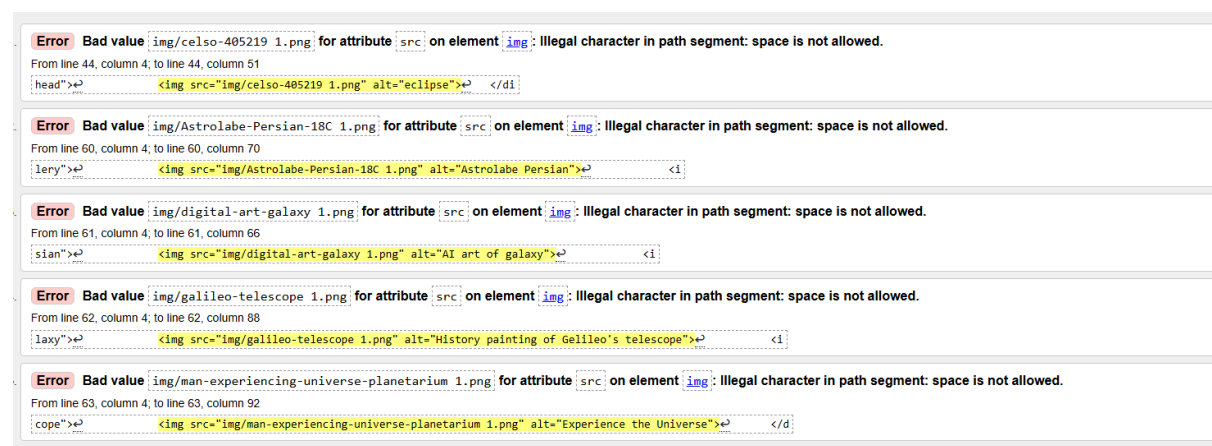


Figure 3: Code validation errors

Further evaluation was performed using **Lighthouse**, a built-in auditing tool in Google Chrome's Developer Tools. Lighthouse measures web quality across four key areas: performance, accessibility, best practices and search-engine optimization (SEO). The final Lighthouse results were as follows:

- Performances: 40-70
- Accessibility: 91
- Best Practices: 96
- SEO: 100



The high score in SEO and Best Practices indicates that the website uses semantic structure, descriptive meta tags, and responsive design effectively. The accessibility scores of 91 shows that most accessibility principles were followed, including proper color contrast, clear heading hierarchy, and alternative text for images. The lower scores on performance were affected by image loading times and file sizes, although images were optimized, some large hero images still caused slower initial load times.

Accessibility testing was conducted both manually and using Lighthouse accessibility audit. The site followed most of the WCAG 2.1 guidelines. All images included descriptive alt text, and headings were ordered logically. Buttons and links provided clear visual feedback on hover and were keyboard navigable. The site's color palette and font size were reviewed to ensure readability.

The website was also tested on different devices to ensure that the layout and images scaled correctly. Media queries worked as intended, with the navigation menu collapsing into a mobile-friendly hamburger icon on smaller screens. The website was tested on Google Chrome and Microsoft Edge to ensure visual consistency and functional stability. The layout and interactive components behaved consistently.

2.3. Conclusion

This project successfully met its main goal of designing and developing a responsive, accessible, and visually appealing website for the Community Science Museum. The process combined creativity with technical problem-solving, resulting in a functional website that effectively represents the museum's purpose.

From the beginning, planning played a crucial role in keeping the project structured and manageable. Using a Kanban board made it easier to organize tasks and stay on track throughout each stage; from planning and design to coding and testing. Dividing work into sections like Backlog, To Do, In Progress, and Done gave a clear overview of progress and helped maintain consistency, much like a real-world development environment. Although some of the tasks may have been finished before planned, the plan was successfully completed.

The design phase focused strongly on user experience (UX) and visual design (UI). Researching other museum and educational websites helped design decisions around



layout, navigation, and content flow. Working in Figma made it possible to experiment with different layouts before coding, which saved time and improved clarity.

The development phase brought the design to life using HTML, CSS and JavaScript. Semantic HTML provided structure and accessibility, while Flexbox were used to create a responsive layout. CSS variables helped maintain consistent styling across pages, and JavaScript added interactivity to responsive navigation menu.

Testing and evaluation confirmed that the website functioned as intended.

Validation using the W3C Markup Validation Service showed that the code followed web standards, with only minor adjustments needed for image file names that contained spaces. Lighthouse analysis provided further insight into technical quality with strong results: 100 for SEO, 96 for Best Practices, 91 for Accessibility and 40-50 for Performance. These results show that the site is well-built and accessible, though there is room for improvement in performance through further optimization.

Overall, this project has been an excellent learning experience that connected design thinking with real-world frontend development. It strengthened skills in responsive design, accessibility, and project planning, and provided hands-on experience with tools commonly used in the industry.



3. References

Books and articles:

Robbins, J.N. (2018): *Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics, 5th Edition*. O'Reilly Media Incl.

Kirpatrick A., O Connor J., Campbell A., Cooper M (2025). *Web Content Accessibility Guidelines (WCAG) 2.1*. <https://www.w3.org/TR/WCAG21/#non-text-content>

Youtube:

Mavi Design (2022): *Create a Button a HOVER Functionally in 128 seconds (Figma Tutorial)*. <https://www.youtube.com/watch?v=AHBEpMD2dZo>

LiveCode (2022): *Making a Responsive Navbar in HTML & CSS – Explained from A to Z with Menu Hamburger*. <https://www.youtube.com/watch?v=HQopEEurQYE>

Images (external used images, not included in provided folder):

Teacher:

- TbeP (2019). The Tampa Bay Estuary Program. <https://unsplash.com/photos/man-standing-in-front-of-women-lcUQNzaHrvq>

Cosmology:

- Andrew Dunn. Persian astrolabe from 18th-century. <https://no.wikipedia.org/wiki/Astrolabium#/media/Fil:Astrolabe-Persian-18C.jpg>
- Freepik. Digital art galaxy. https://www.freepik.com/free-ai-image/digital-art-galaxy_93625853.htm
- Gabriele vanin.it in What is Galileo's Telescope? https://www.universetoday.com/article_images/Galileo_Bertini_fresco_of_Galileo_Galilei_and_Doge_of_Venice-e1446672024312-580x516.jpg
- Freepik. Man experiencing the universe at a planetarium. https://www.freepik.com/free-photo/man-experiencing-universe-planetarium_13461053.htm

Evolution:

- Eugene Zhyvchik (2020). People walking on sidewalk during daytime. <https://unsplash.com/photos/people-walking-on-sidewalk-during-daytime-xJY-7gtC38o>

Robotics and AI:



- Rawpixel.com: AI chip artificial intelligence, future technology innovation. https://www.freepik.com/free-photo/ai-chip-artificial-intelligence-future-technology-innovation_17850485.htm#fromView=search&page=1&position=0&uuid=ab31c218-448e-44a5-9df9-6161ea1ca97f&query=artificial+intelligence
- Freepik. Portrait of robot scifi (AI generate photo). https://www.freepik.com/free-ai-image/portrait-robot-scifi_359212370.htm#fromView=search&page=1&position=46&uuid=da763eec-c4a4-4052-bb98-ac71872e3e4f&query=artificial+intelligence+robot
- Freepik. 3d rendering of biorobots concept. https://www.freepik.com/free-photo/3d-rendering-biorobots-concept_29317036.htm#fromView=keyword&page=1&position=0&uuid=f60d28eb-3a50-410a-b577-42734bf6ed8a&query=Robotics+ai+synergy
- Rawpixel. Robot handshake human background, futuristic digital age. https://www.freepik.com/free-photo/robot-handshake-human-background-futuristic-digital-age_17850426.htm#fromView=search&page=1&position=35&uuid=357dae03-00ae-492f-baca-f85f60400de3&query=robots+shake+hand

Ecology page:

- Photorealistic earth planet, AI Generated photo: https://www.freepik.com/free-ai-image/photorealistic-earth-planet_94075659.htm#fromView=search&page=1&position=23&uuid=a3396421-do86-420e-8ef2-dea4f1acoc14&query=global+warming
- Freepik: Climate change with industrial pollution, https://www.freepik.com/free-photo/climate-change-with-industrial-pollution_21248835.htm#fromView=search&page=1&position=1&uuid=boc3c85f-c9cf-4d68-b49c-549c7ee4beeb&query=pollution
- Jcomp: Environment conservation in the garden for children. https://www.freepik.com/free-photo/environmental-conservation-garden-children_5219379.htm#fromView=search&page=1&position=1&uuid=020d9c98-5a7e-4c54-9445-b98a9ac314a9&query=ecology
- Jason Mavrommatis (2017): Wind turbines on snowy mountain under clear blue sky during daytime. <https://unsplash.com/photos/wind-turbines-on-snowy-mountain-under-clear-blue-sky-during-daytime-nyL-rzwP-Mk>
- Girl with red hat (2020): Polar bear on white sand during daytime. <https://unsplash.com/photos/polar-bear-on-white-sand-during-daytime-hdLjPfcyhyE>



4. Acknowledgements

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5. Appendices

Low fidelity wireframe:

https://www.canva.com/design/DAG5E9cTfvY/LbQVoQP4YJpN_itPeQKPJQ/edit?utm_content=DAG5E9cTfvY&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

High fidelity prototype:

<https://www.figma.com/design/f9UvDh5YhKFoxgqIMeByrZ/SP1-ADA?node-id=12-188&t=MOruXdDJmfCYIYPn-1>

Kanban board: <https://github.com/users/lee-wong13/projects/4/views/1>

Live website: <https://lee-wong13.github.io/Semester-Project-Ada/index.html>

